

Week 16

Keel Bone Chicken Project

12/15/2025



College of Engineering

This Week

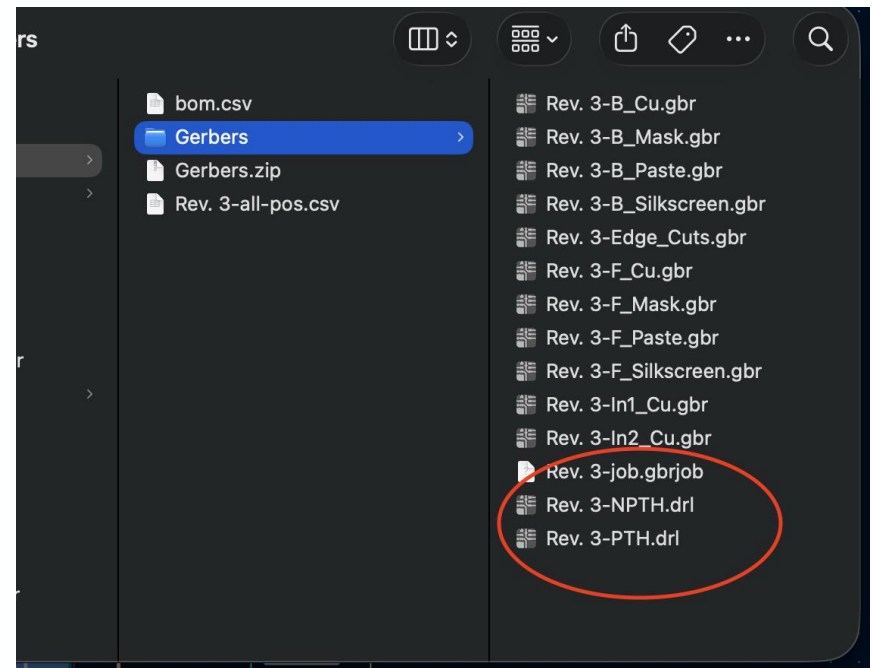
This week's progress

- PCB Regeneration
 - Creation of Drill Files
- Case Study on Data Options
 - (1) Local
 - (2) Cloud
 - (3) Hybrid
 - (4) React Framework
- Rework of Wheatstone Bridge
- Delivery of Device for Data Logging to the Agriculture Department

PCB Regeneration

Creation of Drill Files

- Shipment Delayed
- Contacted JLCPCB to Inquire about the shipment delay
 - Initially a problem with parts sourcing
 - Looked closer and it was an issue with drill file generation
 - Regenerated drill files
 - New Shipment before next week



Case Study on Data Options

Local Storage

- Data stored will be stored on the device or on a laptop running an application
- Highly reliable for offline environments and easily supports four-sensor data
- Possibility to use external hardware to serve as a bridge between device and storage
- Minimal cost, more difficult to develop

Case Study on Data Options

Hybrid Option

- Local logging with periodic cloud sync (3 times x 8 hours / day).
- Reduces bandwidth usage while providing remote accessibility and backup redundancy.
- Cost: moderate—local hardware + limited cloud storage (No need for higher end web options)
- Medium cost, medium difficulty to develop

Case Study on Data Options

Cloud Option

- Data uploaded directly to GitHub, AWS, or Firebase.
- Enables remote monitoring based on reliable server providers

Approximate costs:

GitHub: \$4–\$12/month for private repo tiers with higher storage

AWS S3: \$0.023/GB per month for storage; additional data transfer charges

Firebase: \$25–\$100/month depending on read/write volume and database usage

- Requires stable internet (Not a problem for the agriculture department)
- Higher data uploads would cost more (Unsure if this would be a problem due to the nature of our data)
- Higher cost, easy to develop

Case Study on Data Options

React Framework (For iOS/Android/PMD Platform)

Expo / Expo Go:

- Easiest cross-platform development for Android/iOS; development environment already released to App Store/Google Play Store
- Ideal for quickly testing BLE/USB
- Free, no cloud development is necessarily needed

React Native:

- More customization, better performance a mobile device to obtain data
- Allows integrating native modules if the Nicla requires low-level hardware communication
- Free; No Apple Development account needed for our application

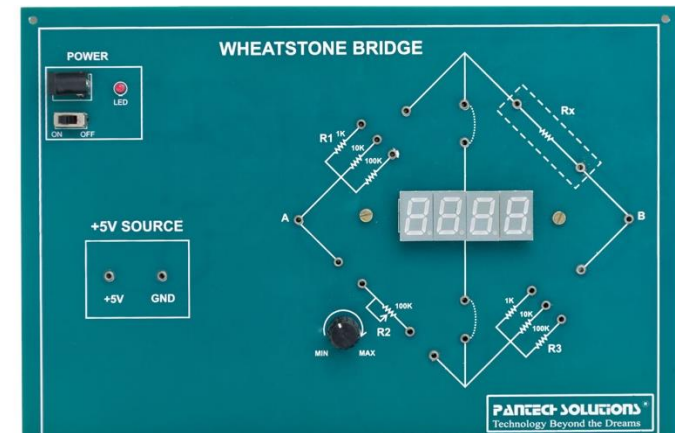
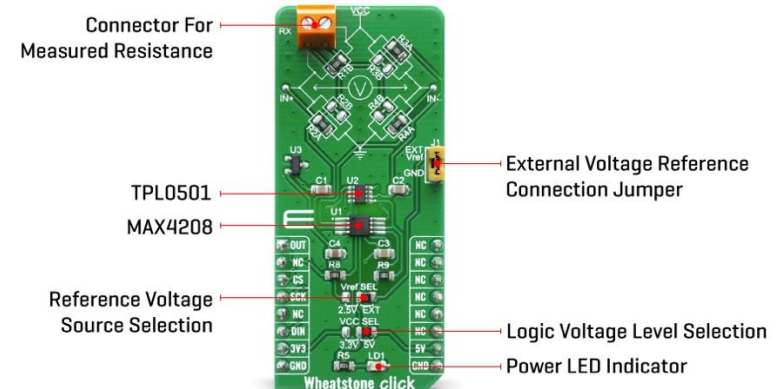
Pure React (Web App):

- Suitable for desktop browser dashboards
- Free; More development time required since all web development occurs on our backend

Rework of Wheatstone Bridge

SMD components would likely be preferable over a chip design

- Used Professor Salas' feedback
- INA333 used better as an amplifier for Wheatstone bridges' output
- SMD components would likely form a better Wheatstone bridge



Delivery of Device for Data Logging to the Agriculture

Showed Executable App, got data working on remote device

- Backpack maybe too small for the Battery + Device (Initial Concern)
- Showed Agriculture Department that the remote app could be compiled
 - Data collection show take place 12/15 to obtain real data from chickens
 - Remote BLE, Battery, Website all work seamlessly

Next Week

Blockers & Next Week's Work

- Ensure PCB shipment arrives
- Get started on data storage option chosen
- Analyze Actual Chicken Data pulled from Agriculture Department
- Finish Rev. 4 of PCB

Blockers:

- Heavy traffic for JLCPCB around this season, need to ensure every component of the order is placed correctly

Thank You

Purdue Polytechnic Institute